

Data Science Project: Movie Industry EDA

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This report summarizes the Exploratory Data Analysis (EDA) conducted on a comprehensive movie dataset to identify trends, audience preferences, and performance metrics.

1. Executive Summary

The objective of this project was to clean, analyze, and visualize movie data. The analysis revealed significant growth in content production over the last decade, with 2021 emerging as a record-breaking year.

2. Key Findings

2021

Peak Release Year

Drama

Most Frequent Genre

Popular

Spider-Man: No Way
Home

The analysis successfully identified that while 'Drama' is the most frequent genre, 'Action' movies often dominate popularity scores. We also established a correlation between popularity metrics and vote counts, filtering out low-engagement data to identify high-quality, high-popularity content.

3. Methodology

- **Data Cleaning:** Handled missing values and standardized release date formatting.
- **Feature Engineering:** Extracted 'Release Year' from raw date timestamps.
- **Statistical Analysis:** Utilized grouping and aggregation to identify genre frequency.
- **Visualization:** Implemented Seaborn/Matplotlib for trend line and bar charts.

4. Conclusion

The findings indicate that the content landscape has shifted significantly towards diversified, high-volume production. This project validates the ability to handle large-scale datasets, filter for quality, and visualize trends effectively for actionable business insights.